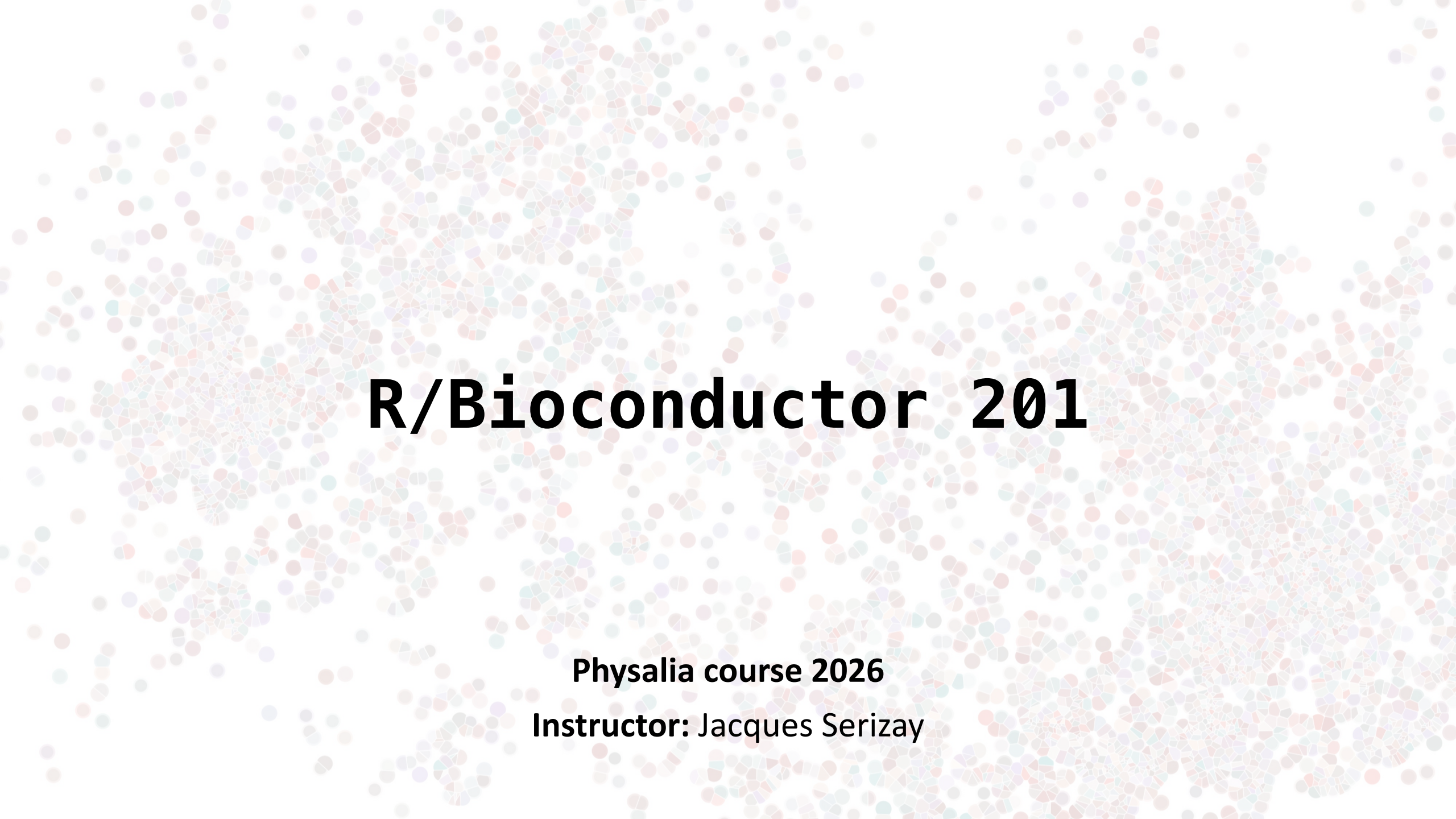




R/Bioconductor 201

Physalia course 2026

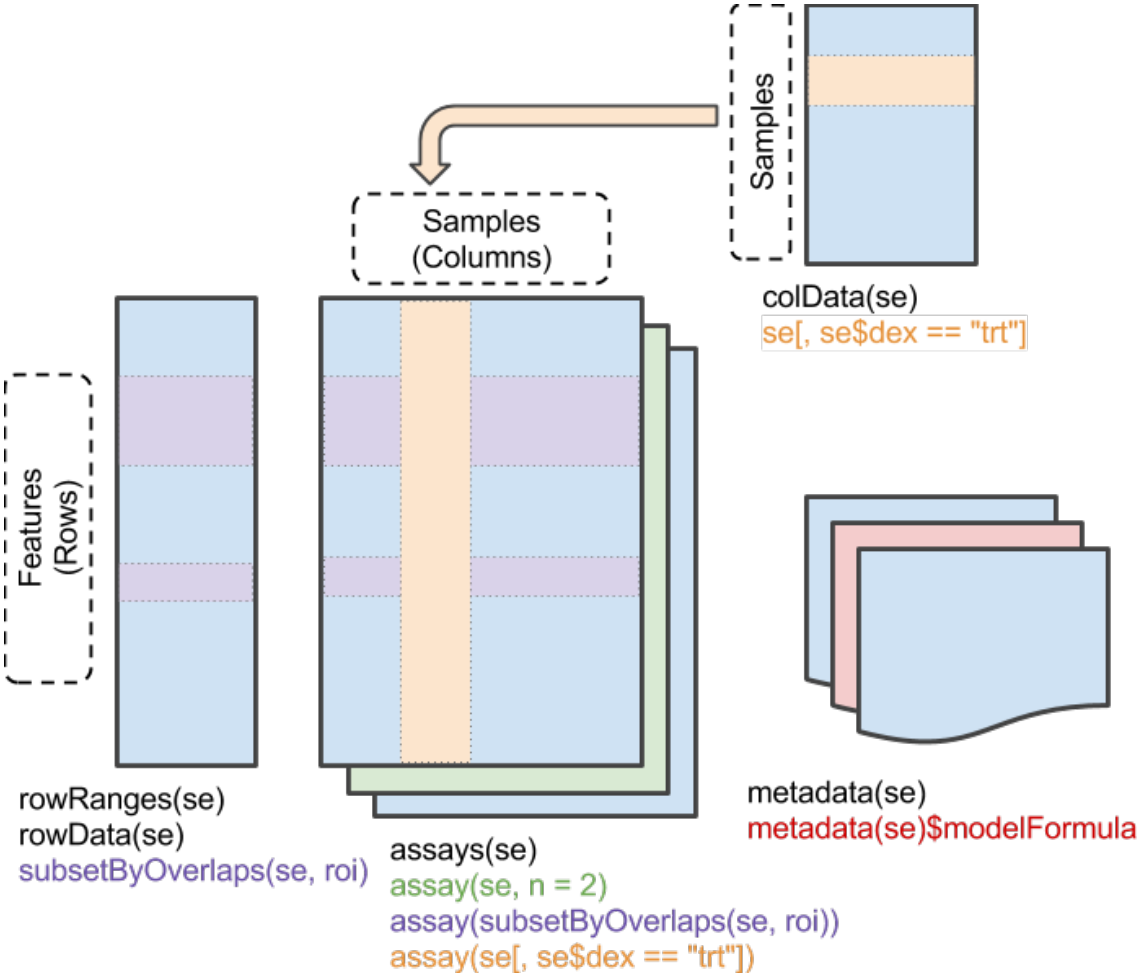
Instructor: Jacques Serizay

Published: 29 January 2015

Orchestrating high-throughput genomic analysis with Bioconductor

Wolfgang Huber , Vincent J Carey, Robert Gentleman, Simon Anders, Marc Carlson, Benilton S Carvalho, Hector Corrada Bravo, Sean Davis, Laurent Gatto, Thomas Girke, Raphael Gottardo, Florian Hahne, Kasper D Hansen, Rafael A Irizarry, Michael Lawrence, Michael I Love, James MacDonald, Valerie Obenchain, Andrzej K Oleś, Hervé Pagès, Alejandro Reyes, Paul Shannon, Gordon K Smyth, Dan Tenenbaum, Levi Waldron & Martin Morgan [-Show fewer authors](#)

Nature Methods **12**, 115–121(2015) | [Cite this article](#)



SummarizedExperiment

Published: 29 January 2015

Orchestrating high-throughput genomic analysis with Bioconductor

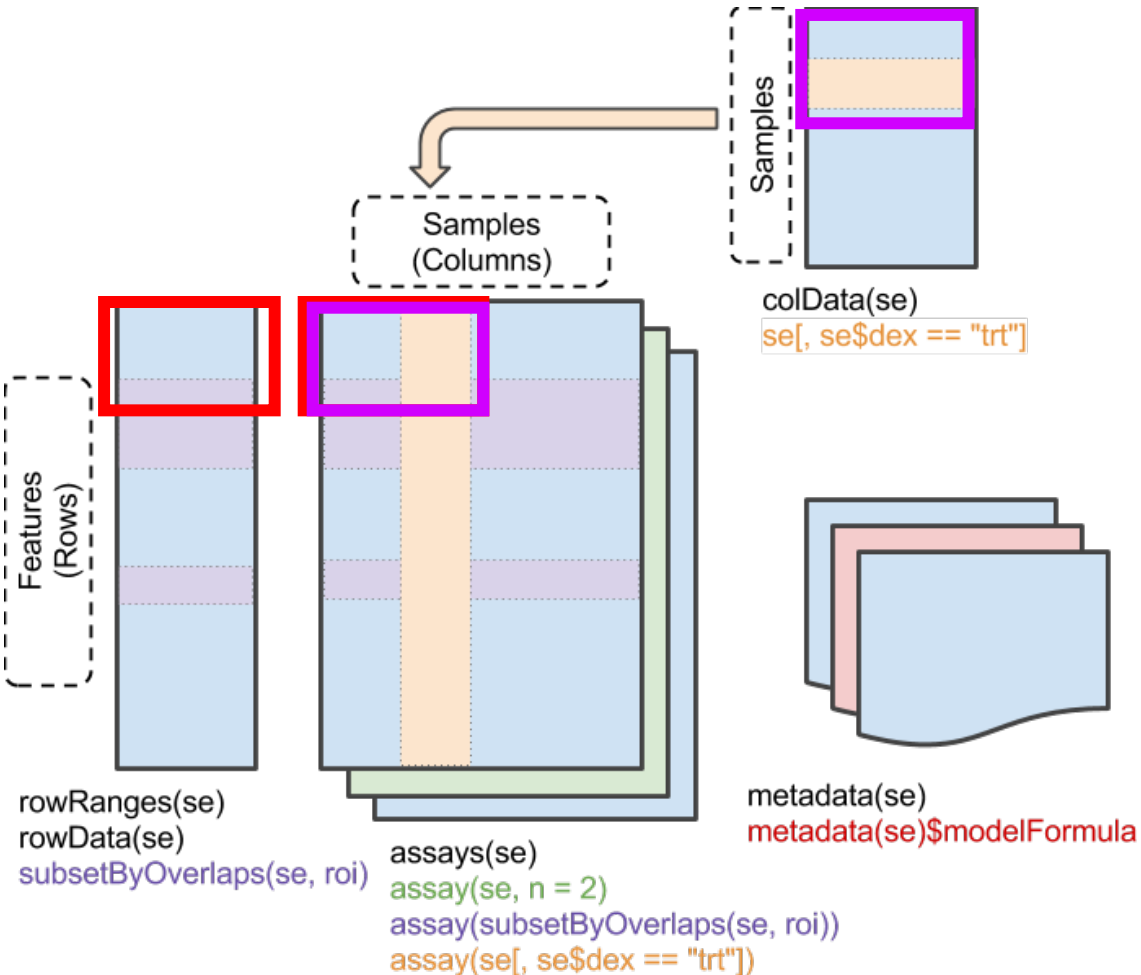
Wolfgang Huber , Vincent J Carey, Robert Gentleman, Simon Anders, Marc Carlson, Benilton S
Carvalho, Hector Corrada Bravo, Sean Davis, Laurent Gatto, Thomas Girke, Raphael Gottardo,
Florian Hahne, Kasper D Hansen, Rafael A Irizarry, Michael Lawrence, Michael I Love, James
MacDonald, Valerie Obenchain, Andrzej K Oleś, Hervé Pagès, Alejandro Reyes, Paul Shannon,
Gordon K Smyth, Dan Tenenbaum, Levi Waldron & Martin Morgan [-Show fewer authors](#)

Nature Methods **12**, 115–121(2015) | [Cite this article](#)

Se [1:10 ,]

Se [, 1:4]

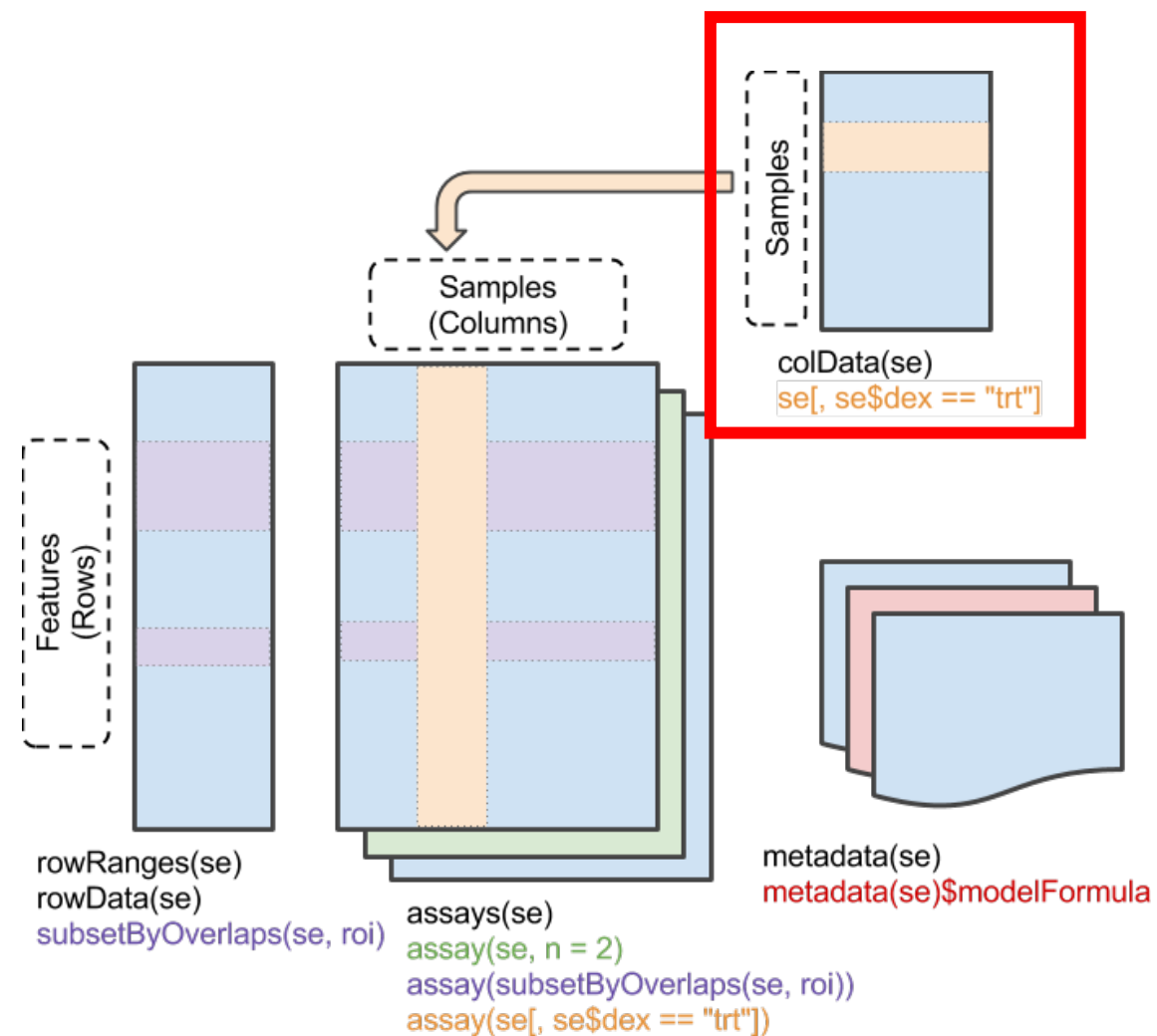
Se [1:10 , 1:4]



SummarizedExperiment

- `colData()`: Annotations on each column, as a `DataFrame`.

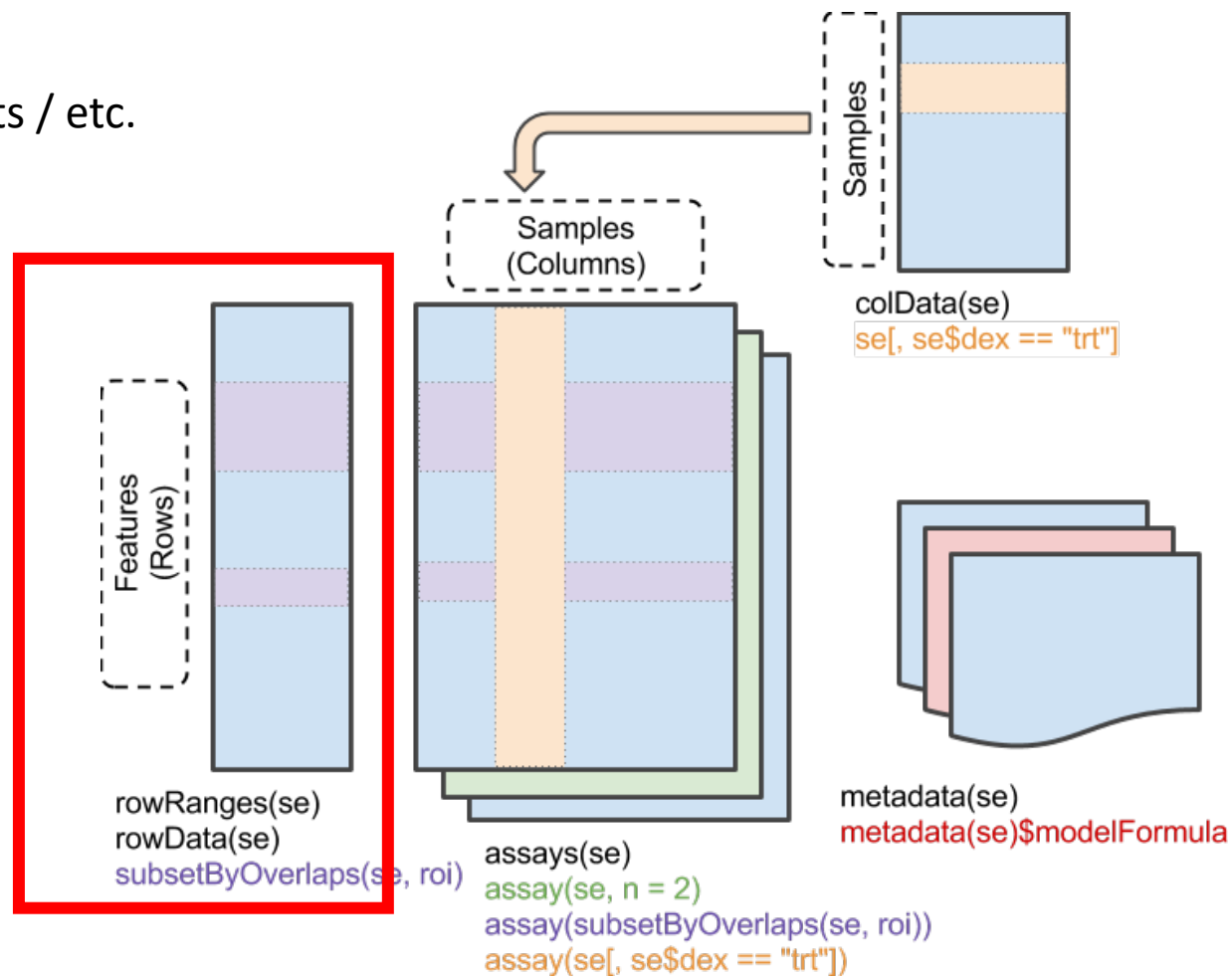
E.g., description of each sample



SummarizedExperiment

- rowData/rowRanges(): Annotations on each row.

E.g., coordinates of gene / exons / peaks in transcripts / etc.



SummarizedExperiment

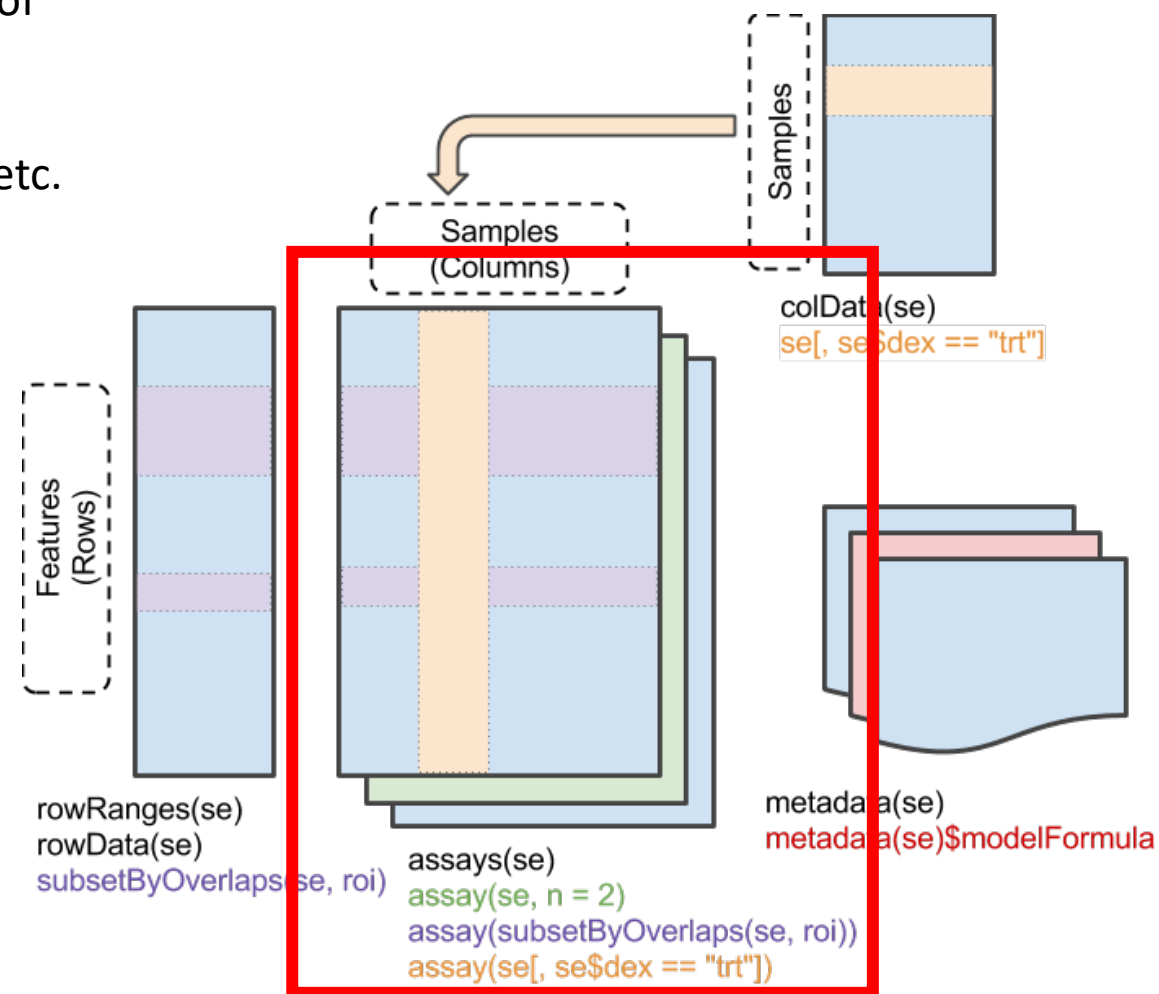
- assay(), assays(): A matrix-like or list of matrix-like objects of identical dimension

rows: refer to **rowRanges**: genes, genomic coordinates, etc.

columns: refer to **colData**: samples, cells, etc.

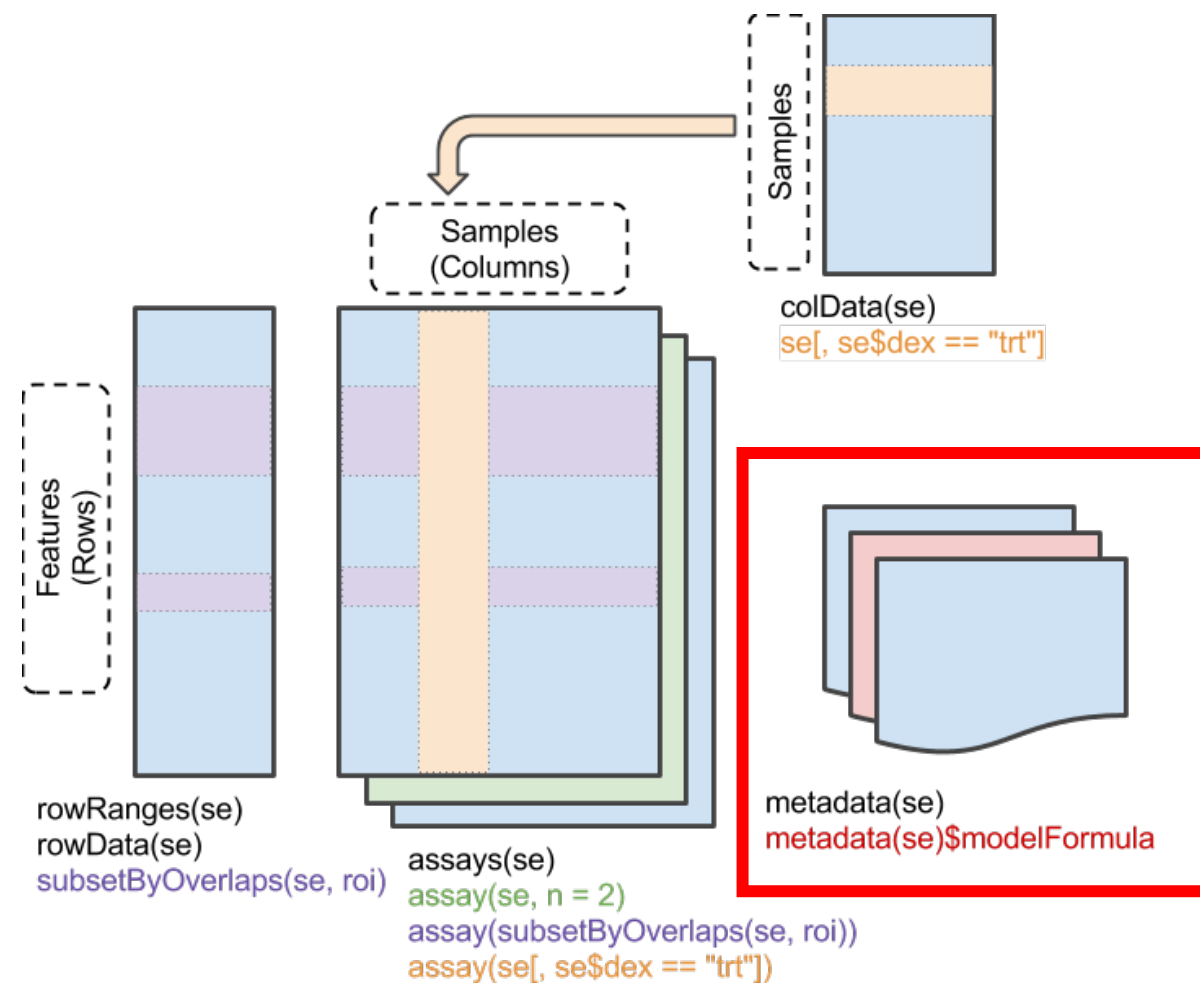
Implements dim(), dimnames() and 2-dimensional [,]

Can be several assays!!!



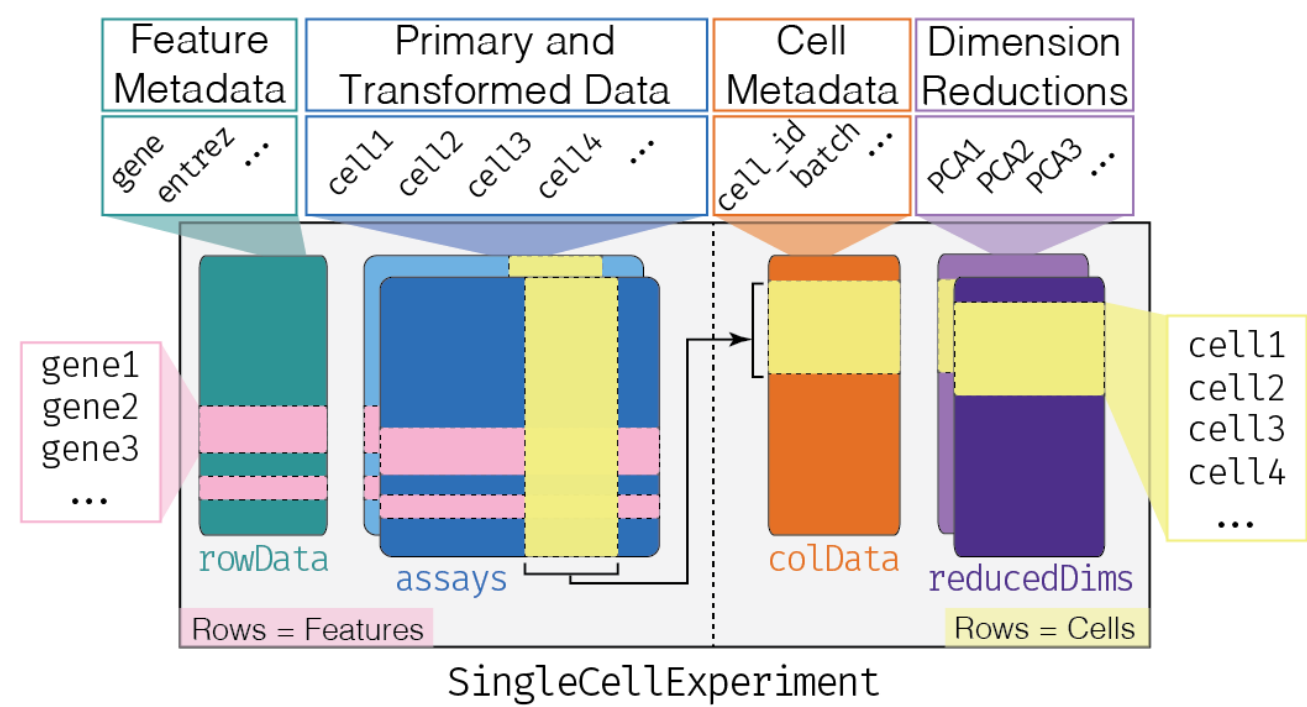
SummarizedExperiment

- `metadata()`: List of unstructured metadata describing the overall content of the object.



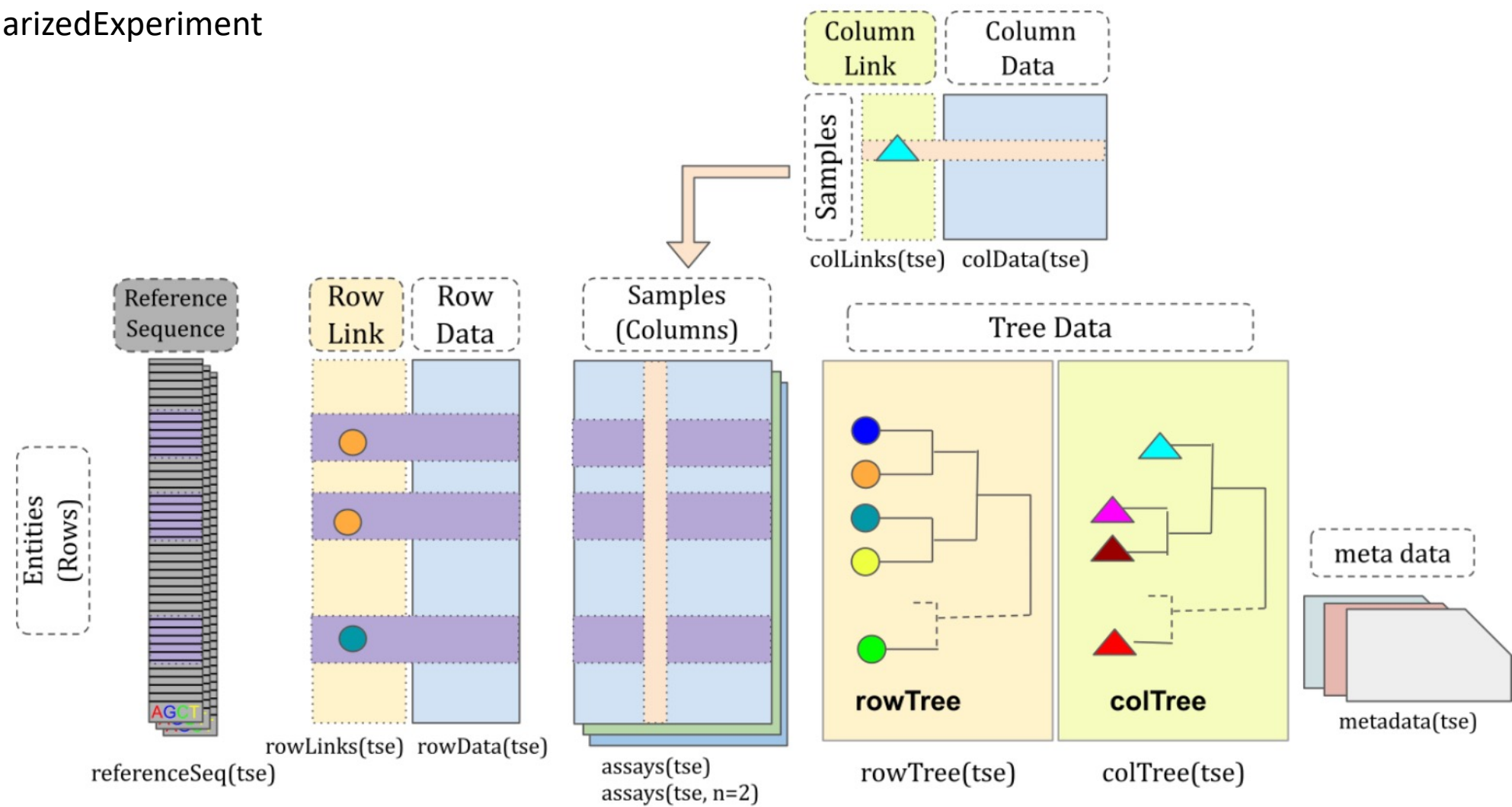
Classes derived from SummarizedExperiment

- SingleCellExperiment



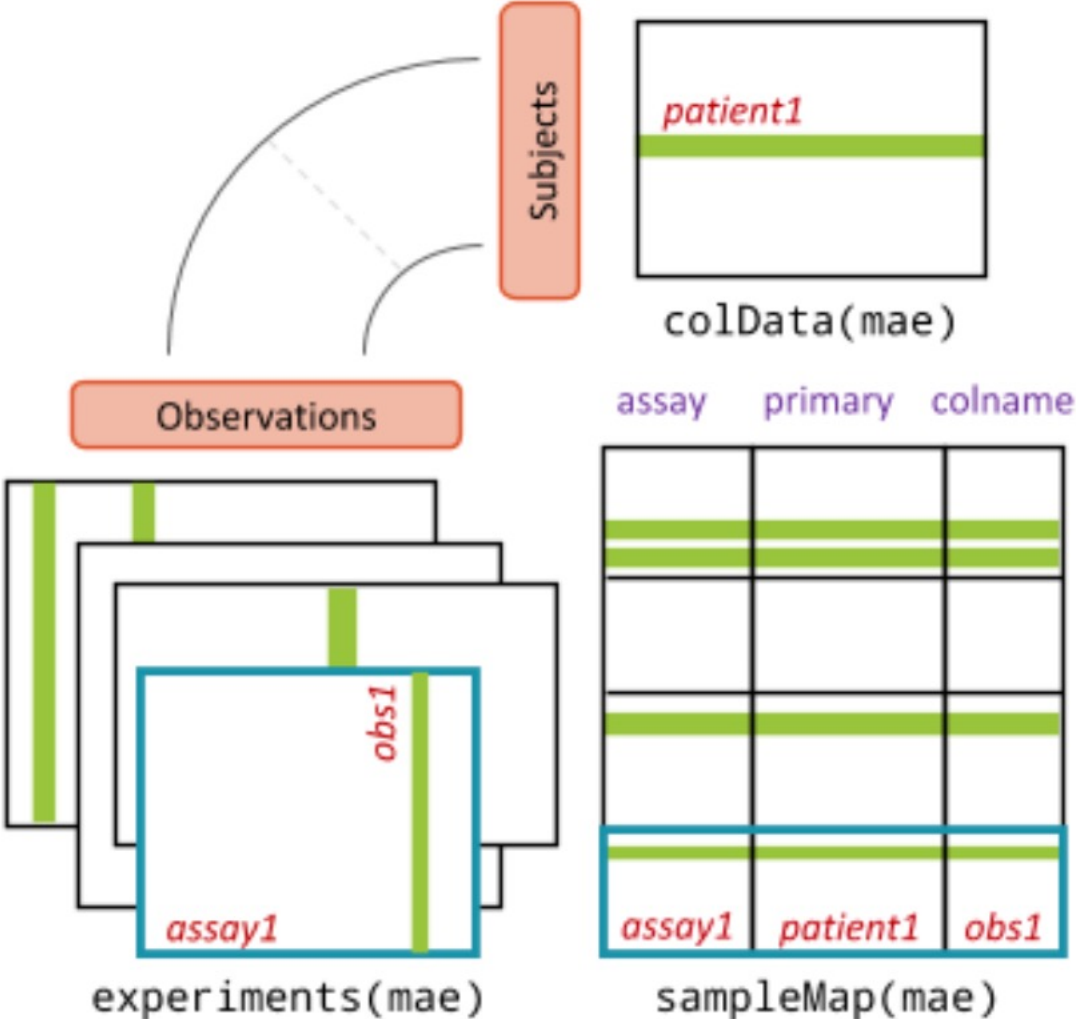
Classes derived from SummarizedExperiment

- TreeSummarizedExperiment



Classes derived from SummarizedExperiment

- MultiAssayExperiment



Genomic analyses require heavy resources

Generally, benefits from parallelization

Genomic analyses require heavy resources

Generally, benefits from parallelization

BiocParallel is a Bioconductor package designed to reduce the complexity faced when developing and using software that performs parallel computations

BiocParallel aims to provide a unified interface to existing parallel infrastructures where code can be easily executed in different environments

Declaring configurations:

registered()

bpparam()

register(..., default = TRUE)

```
> BiocParallel::registered()
$MulticoreParam
class: MulticoreParam
  bpisup: FALSE; bpnworkers: 6; bptasks: 0; bpjobname: BPJOB
  bplog: FALSE; bpthreshold: INFO; bpstopOnError: TRUE
  bpRNGseed: ; bptimeout: 2592000; bpprogressbar: FALSE
  bpexportglobals: TRUE
  bplogdir: NA
  bpresultdir: NA
  cluster type: FORK
```

```
$SnowParam
class: SnowParam
  bpisup: FALSE; bpnworkers: 6; bptasks: 0; bpjobname: BPJOB
  bplog: FALSE; bpthreshold: INFO; bpstopOnError: TRUE
  bpRNGseed: ; bptimeout: 2592000; bpprogressbar: FALSE
  bpexportglobals: TRUE
  bplogdir: NA
  bpresultdir: NA
  cluster type: SOCK
```

```
$SerialParam
class: SerialParam
  bpisup: FALSE; bpnworkers: 1; bptasks: 0; bpjobname: BPJOB
  bplog: FALSE; bpthreshold: INFO; bpstopOnError: TRUE
  bpRNGseed: ; bptimeout: 2592000; bpprogressbar: FALSE
  bpexportglobals: TRUE
  bplogdir: NA
  bpresultdir: NA
```

```
> BiocParallel::bpparam()
class: MulticoreParam
  bpisup: FALSE; bpnworkers: 6; bptasks: 0; bpjobname: BPJOB
  bplog: FALSE; bpthreshold: INFO; bpstopOnError: TRUE
  bpRNGseed: ; bptimeout: 2592000; bpprogressbar: FALSE
  bpexportglobals: TRUE
  bplogdir: NA
  bpresultdir: NA
  cluster type: FORK
```

Declaring configurations:

registered()

bpparam()

register(..., default = TRUE)

MulticoreParam()

SerialParam()

SnowParam()

```
> BiocParallel::registered()
$MulticoreParam
class: MulticoreParam
 bpisup: FALSE; bpnworkers: 6; bptasks: 0; bpjobname: BPJOB
 bplog: FALSE; bpthreshold: INFO; bpstopOnError: TRUE
 bpRNGseed: ; bptimeout: 2592000; bpprogressbar: FALSE
 bpexportglobals: TRUE
 bplogdir: NA
 bpresultdir: NA
 cluster type: FORK

$SnowParam
class: SnowParam
 bpisup: FALSE; bpnworkers: 6; bptasks: 0; bpjobname: BPJOB
 bplog: FALSE; bpthreshold: INFO; bpstopOnError: TRUE
 bpRNGseed: ; bptimeout: 2592000; bpprogressbar: FALSE
 bpexportglobals: TRUE
 bplogdir: NA
 bpresultdir: NA
 cluster type: SOCK

$SerialParam
class: SerialParam
 bpisup: FALSE; bpnworkers: 1; bptasks: 0; bpjobname: BPJOB
 bplog: FALSE; bpthreshold: INFO; bpstopOnError: TRUE
 bpRNGseed: ; bptimeout: 2592000; bpprogressbar: FALSE
 bpexportglobals: TRUE
 bplogdir: NA
 bpresultdir: NA

> BiocParallel::bpparam()
class: MulticoreParam
 bpisup: FALSE; bpnworkers: 6; bptasks: 0; bpjobname: BPJOB
 bplog: FALSE; bpthreshold: INFO; bpstopOnError: TRUE
 bpRNGseed: ; bptimeout: 2592000; bpprogressbar: FALSE
 bpexportglobals: TRUE
 bplogdir: NA
 bpresultdir: NA
 cluster type: FORK
```

Execute in parallel:

bplapply() e.g., bplapply(1:4, FUN)

```
> BiocParallel::registered()
$MulticoreParam
class: MulticoreParam
  bpisup: FALSE; bpnworkers: 6; bptasks: 0; bpjobname: BPJOB
  bplog: FALSE; bpthreshold: INFO; bpstopOnError: TRUE
  bpRNGseed: ; bptimeout: 2592000; bpprogressbar: FALSE
  bpexportglobals: TRUE
  bplogdir: NA
  bpresultdir: NA
  cluster type: FORK
```

```
$SnowParam
class: SnowParam
  bpisup: FALSE; bpnworkers: 6; bptasks: 0; bpjobname: BPJOB
  bplog: FALSE; bpthreshold: INFO; bpstopOnError: TRUE
  bpRNGseed: ; bptimeout: 2592000; bpprogressbar: FALSE
  bpexportglobals: TRUE
  bplogdir: NA
  bpresultdir: NA
  cluster type: SOCK
```

```
$SerialParam
class: SerialParam
  bpisup: FALSE; bpnworkers: 1; bptasks: 0; bpjobname: BPJOB
  bplog: FALSE; bpthreshold: INFO; bpstopOnError: TRUE
  bpRNGseed: ; bptimeout: 2592000; bpprogressbar: FALSE
  bpexportglobals: TRUE
  bplogdir: NA
  bpresultdir: NA
```

```
> BiocParallel::bpparam()
class: MulticoreParam
  bpisup: FALSE; bpnworkers: 6; bptasks: 0; bpjobname: BPJOB
  bplog: FALSE; bpthreshold: INFO; bpstopOnError: TRUE
  bpRNGseed: ; bptimeout: 2592000; bpprogressbar: FALSE
  bpexportglobals: TRUE
  bplogdir: NA
  bpresultdir: NA
  cluster type: FORK
```

Bioconductor is not only for analysis packages

<https://www.bioconductor.org/packages/release/BiocViews.html>

- ▼ Software (1974)
 - ▶ AssayDomain (791)
 - ▶ BiologicalQuestion (822)
 - ▶ Infrastructure (456)
 - ▶ ResearchField (902)
 - ▶ StatisticalMethod (727)
 - ▶ Technology (1251)
 - ▶ WorkflowStep (1081)

Bioconductor is not only for analysis packages

<https://www.bioconductor.org/packages/release/BiocViews.html>

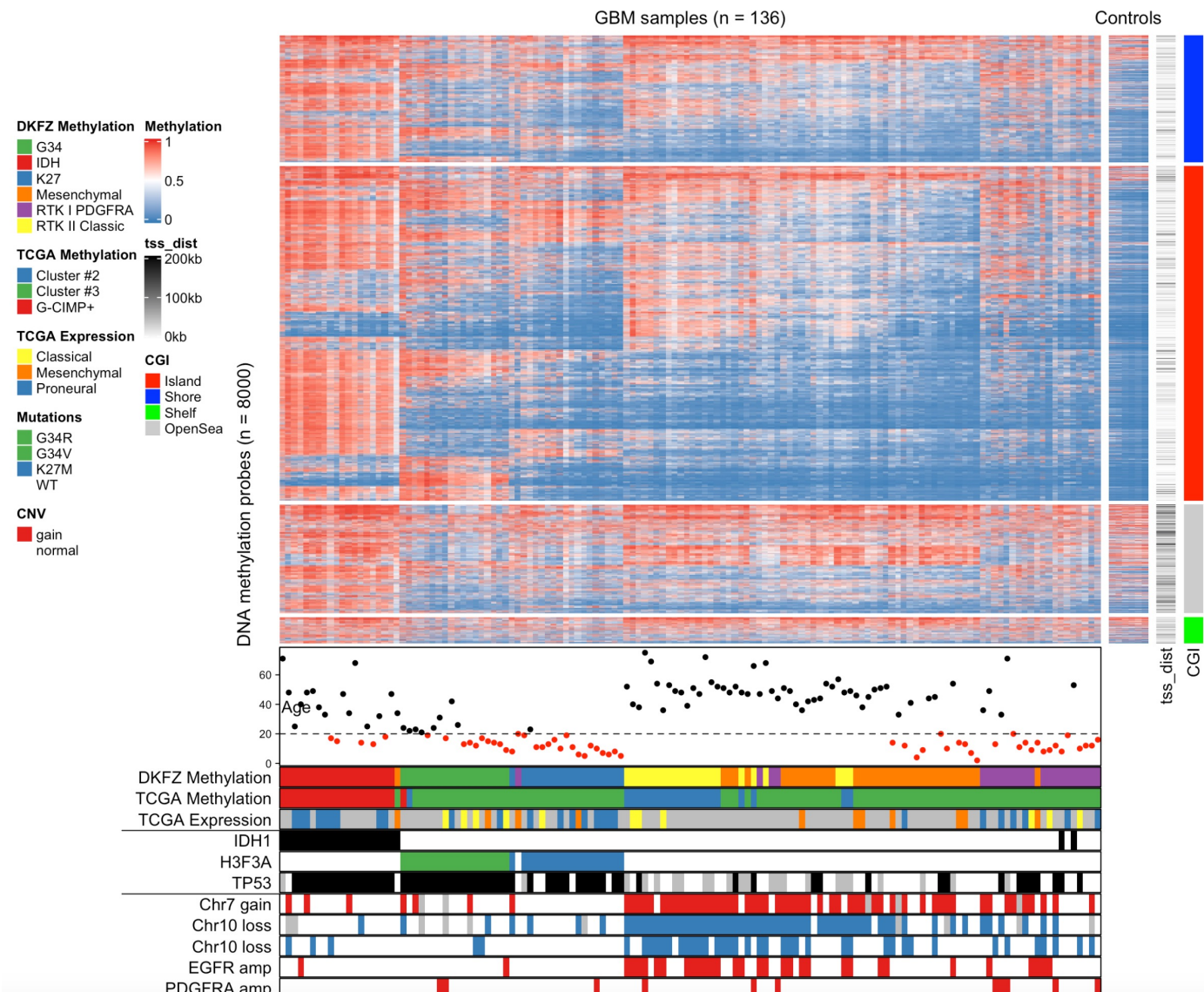
- [Software \(1974\)](#)
 - [AssayDomain \(791\)](#)
 - [BiologicalQuestion \(822\)](#)
 - [Infrastructure \(456\)](#)
 - [ResearchField \(902\)](#)
 - [StatisticalMethod \(727\)](#)
 - [Technology \(1251\)](#)
 - [WorkflowStep \(1081\)](#)
- [ExperimentData \(398\)](#)
 - [AssayDomainData \(81\)](#)
 - [DiseaseModel \(90\)](#)
 - [OrganismData \(139\)](#)
 - [PackageTypeData \(41\)](#)
 - [RepositoryData \(94\)](#)
 - [ReproducibleResearch \(22\)](#)
 - [SpecimenSource \(103\)](#)
 - [TechnologyData \(266\)](#)
- [AnnotationData \(971\)](#)
 - [ChipManufacturer \(388\)](#)
 - [ChipName \(196\)](#)
 - [CustomArray \(2\)](#)
 - [CustomDBSchema \(6\)](#)
 - [FunctionalAnnotation \(31\)](#)
 - [Organism \(634\)](#)
 - [PackageType \(682\)](#)
 - [SequenceAnnotation \(1\)](#)
- [Workflow \(28\)](#)
 - [AnnotationWorkflow \(3\)](#)
 - [BasicWorkflow \(5\)](#)
 - [EpigeneticsWorkflow \(4\)](#)
 - [GeneExpressionWorkflow \(11\)](#)
 - [GenomicVariantsWorkflow \(2\)](#)
 - [ImmunoOncologyWorkflow \(14\)](#)
 - [ProteomicsWorkflow \(2\)](#)
 - [ResourceQueryingWorkflow \(2\)](#)
 - [SingleCellWorkflow \(2\)](#)

Complex heatmaps reveal patterns and correlations in multidimensional genomic data FREE

Zuguang Gu, Roland Eils, Matthias Schlesner ✉ Author Notes

Bioinformatics, Volume 32, Issue 18, 15 September 2016, Pages 2847–2849,
<https://doi.org/10.1093/bioinformatics/btw313>

Published: 20 May 2016 Article history ▾



- ▼ Software (1974)
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 - ▶ StatisticalMethod (727)
 - ▶ Technology (1251)
 - ▶ WorkflowStep (1081)

systemPipeR

platforms

all

rank

148 / 1974

posts

0

in Bioc

6 years

build

ok

updated

< 3 months

dependencies

154

DOI: [10.18129/B9.bioc.systemPipeR](https://doi.org/10.18129/B9.bioc.systemPipeR)  

systemPipeR: NGS workflow and report generation environment

Bioconductor version: Release (3.12)

R package for building and running automated end-to-end analysis workflows for a wide range of next generation sequence (NGS) applications such as RNA-Seq, ChIP-Seq, VAR-Seq and Ribo-Seq. Important features include a uniform workflow interface across different NGS applications, automated report generation, and support for running both R and command-line software, such as NGS aligners or peak/variant callers, on local computers or compute clusters. Efficient handling of complex sample sets and experimental designs is facilitated by a consistently implemented sample annotation infrastructure. Instructions for using systemPipeR are given in the Overview Vignette (HTML). The remaining Vignettes, linked below, are workflow templates for common NGS use cases.

Reporting tools

▼ Software (1974)

- ▶ AssayDomain (791)
- ▶ BiologicalQuestion (822)
- ▶ Infrastructure (456)
- ▶ ResearchField (902)
- ▶ StatisticalMethod (727)
- ▶ Technology (1251)
- ▶ WorkflowStep (1081)

```
DESeq2Report::DESeq2Report(  
  dds = dds,  
  project = "OsmoResponse",  
  intgroup = "timepoint"  
)
```

Bioconductor is not only for analysis packages

<https://www.bioconductor.org/packages/release/BiocViews.html>

- ▶ Software (1974)
- ▶ AnnotationData (971)
- ▼ ExperimentData (398)
 - ▶ AssayDomainData (81)
 - ▶ DiseaseModel (90)
 - ▶ OrganismData (139)
 - ▶ PackageTypeData (41)
 - ▶ RepositoryData (94)
 - ReproducibleResearch (22)
 - ▶ SpecimenSource (103)
 - ▶ TechnologyData (266)
- ▶ Workflow (28)

Retrieving specific experiments

RangedSummarizedExperiment for time course RNA-Seq of fission yeast in response to stress, by Leong et al., Nat Commun 2014.

Bioconductor version: Release (3.12)

This package provides a RangedSummarizedExperiment object of read counts in genes for a time course RNA-Seq experiment of fission yeast (*Schizosaccharomyces pombe*) in response to oxidative stress (1M sorbitol treatment) at 0, 15, 30, 60, 120 and 180 mins. The samples are further divided between a wild-type group and a group with deletion of *atf21*. The read count matrix was prepared and provided by the author of the study: Leong HS, Dawson K, Wirth C, Li Y, Connolly Y, Smith DL, Wilkinson CR, Miller CJ. "A global non-coding RNA system modulates fission yeast protein levels in response to stress". *Nat Commun* 2014 May 23;5:3947. PMID: 24853205. GEO: GSE56761.

Author: Michael Love

Maintainer: Michael Love <michaelisaiahlove at gmail.com>

Citation (from within R, enter `citation("fission")`):

Leong, S. H, Dawson, K., Wirth, C., Li, Y., Connolly, Y., Smith, L. D, Wilkinson, R. C, Miller, J. C (2014). "A global non-coding RNA system modulates fission yeast protein levels in response to stress." *Nat Commun*, **5**, 3947. <http://www.ncbi.nlm.nih.gov/pubmed/24853205>.

```
> library(fission)
> fission
class: RangedSummarizedExperiment
dim: 7039 36
metadata(1): ''
assays(1): counts
rownames(7039): SPAC212.11 SPAC212.09c ... SPMITTRNAGLU.01 SPMIT.11
rowData names(2): symbol biotype
colnames(36): GSM1368273 GSM1368274 ... GSM1368307 GSM1368308
colData names(4): strain minute replicate id
> rowRanges(fission)
GRanges object with 7039 ranges and 2 metadata columns:
      seqnames      ranges strand |      symbol      biotype
      <Rle>      <IRanges> <Rle> | <character> <factor>
SPAC212.11      I      1-5662   - |      tlh1 protein_coding
SPAC212.09c      I    7619-9274   + | SPAC212.09c pseudogene
SPNCRNA.70      I 11027-11556   - | SPNCRNA.70 ncRNA
SPAC212.12      I 15855-16226   + | SPAC212.12 protein_coding
SPAC212.04c      I 21381-23050   + | SPAC212.04c protein_coding
...
SPMITTRNATYR.01 MT 17257-17342   + | SPMITTRNATYR.01 tRNA
SPMITTRNAILE.02 MT 17542-17613   + | SPMITTRNAILE.02 tRNA
SPMIT.10        MT 17806-18030   + |      atp9 protein_coding
SPMITTRNAGLU.01 MT 18404-18475   + | SPMITTRNAGLU.01 tRNA
SPMIT.11        MT 18561-19307   + |      cox2 protein_coding
-----
seqinfo: 4 sequences from an unspecified genome; no seqlengths
```

Retrieving specific experiments

```
library(VariantAnnotation)
vcf <- readVcf(
  system.file("extdata", "SonVariantsChr21.vcf.gz", package = "AshkenazimSonChr21"),
  genome = "hg19"
)
info(vcf)
# A tibble: 94,527 x 35
  AC      AF      AN      DP      QD BLOCKAVG_min30p... BaseQRankSum DS      Dels      END      FS      HRun HaplotypeScore
  <I<l> <chr> <int> <int> <dbl> <lgl> <dbl> <lgl> <dbl> <int> <dbl> <int> <dbl>
1 <int... 0.50      2     38  8.25 FALSE -0.923 FALSE      0     NA      0      0      1.98
2 <int... 0.50      2     37 19.5  FALSE -0.334 FALSE      0     NA     1.44      1      1.00
3 <int... 0.50      2     49 23.0  FALSE -0.683 FALSE      0     NA    11.8      1      0.867
4 <int... 0.50      2     62 20.0  FALSE  1.40  FALSE      0     NA     1.00      0      0
5 <int... 0.50      2     57 10.8  FALSE -1.44  FALSE      0     NA      0      0      0
6 <int... 0.50      2     56 10.8  FALSE -1.46  FALSE      0     NA      0      1     12.0
7 <int... 0.50      2     55  7.13 FALSE -0.141 FALSE      0     NA      0      0     14.0
8 <int... 0.50      2     50 16.8  FALSE  0.842 FALSE      0     NA      0      0      0
9 <int... 0.50      2     73 18.0  FALSE  0.456 FALSE      0     NA     9.32      2     0.789
10 <int... 0.50      2     86  8.44 FALSE -0.005 FALSE      0     NA      0      2     5.86
# ... with 94,517 more rows, and 22 more variables: InbreedingCoeff <dbl>, MQ <dbl>, MQ0 <int>, MQRankSum <dbl>,
# ReadPosRankSum <dbl>, SB <dbl>, VQSL0D <dbl>, culprit <chr>, set <chr>, CSQT <I<list>>, CSQR <I<list>>, AA <chr>,
# GMAF <I<list>>, EVS <I<list>>, cosmic <I<list>>, clinvar <I<list>>, phastCons <lgl>, Variant.type <I<list>>,
# Gene.name <I<list>>, Gene.component <I<list>>, phyloP <dbl>, SNP.Frequency <dbl>
```

Retrieving specific experiments

```
> scRNAseq::listDatasets()
DataFrame with 46 rows and 5 columns
```

	Reference	Taxonomy	Part	Number	Call
	<character>	<integer>	<character>	<integer>	<character>
1	@aztekin2019identifi..	8355	tail	13199	AztekinTailData()
2	@bach2017differentia..	10090	mammary gland	25806	BachMammaryData()
3	@baron2016singlecell	9606	pancreas	8569	BaronPancreasData('h..
4	@baron2016singlecell	10090	pancreas	1886	BaronPancreasData('m..
5	@buettner2015computa..	10090	embryonic stem cells	288	BuettnerESData()
...	...	...	...	...	...
42	@wu2019advantages	10090	kidney	17542	WuKidneyData()
43	@xin2016rna	9606	pancreas	1600	XinPancreasData()
44	@zeisel2015brain	10090	brain	3005	ZeiselBrainData()
45	@zilionis2019singlec..	9606	lung	173954	ZilionisLungData()
46	@zilionis2019singlec..	10090	lung	17549	ZilionisLungData('mo..

```
> ZeiselBrainData()
snapshotDate(): 2020-10-02
see ?scRNAseq and browseVignettes('scRNAseq') for documentation
loading from cache
see ?scRNAseq and browseVignettes('scRNAseq') for documentation
loading from cache
see ?scRNAseq and browseVignettes('scRNAseq') for documentation
loading from cache
snapshotDate(): 2020-10-02
see ?scRNAseq and browseVignettes('scRNAseq') for documentation
loading from cache
class: SingleCellExperiment
dim: 20006 3005
metadata(0):
assays(1): counts
rownames(20006): Tspan12 Tshz1 ... mt-Rnr1 mt-Nd4l
rowData names(1): featureType
colnames(3005): 1772071015_C02 1772071017_G12 ... 1772066098_A12 1772058148_F03
colData names(10): tissue group # ... level1class level2class
reducedDimNames(0):
altExpNames(2): ERCC repeat
```

Bioconductor Annotation packages

	Package	Maintainer	Title
BS	BSgenome.Scerevisiae.UCSC.sacCer3	Bioconductor Package Maintainer	Saccharomyces cerevisiae (Yeast) full genome (UCSC version sacCer3)
BS	BSgenome.Scerevisiae.UCSC.sacCer2	Bioconductor Package Maintainer	Saccharomyces cerevisiae (Yeast) full genome (UCSC version sacCer2)
TxDb	TxDb.Scerevisiae.UCSC.sacCer3.sgdGene	Bioconductor Package Maintainer	Annotation package for TxDb object(s)
BS	BSgenome.Scerevisiae.UCSC.sacCer1	Bioconductor Package Maintainer	Saccharomyces cerevisiae (Yeast) full genome (UCSC version sacCer1)
	hom.Sc.inp.db	Bioconductor Package Maintainer	Homology information for Saccharomyces cerevisiae from Inparanoid
	MeSH.Sce.S288c.eg.db	Koki Tsuyuzaki	Mapping table for Saccharomyces cerevisiae S288c Gene ID to MeSH
TxDb	TxDb.Scerevisiae.UCSC.sacCer2.sgdGene	Bioconductor Package Maintainer	Annotation package for TxDb object(s)
org	org.Sc.sgd.db	Bioconductor Package Maintainer	Genome wide annotation for Yeast

```
> library(org.Sc.sgd.db)
> org.Sc.sgd.db
OrgDb object:
 | DBSCHEMAVERSION: 2.1
 | Db type: OrgDb
 | Supporting package: AnnotationDbi
 | DBSCHEMA: YEAST_DB
 | ORGANISM: Saccharomyces cerevisiae
 | SPECIES: Yeast
 | YGSOURCENAME: Yeast Genome
 | YGSOURCEURL: http://sgd-archive.yeastgenome.org
 | YGSOURCEDATE: 2019-Oct25
 | CENTRALID: ORF
 | TAXID: 559292
 | KEGGSOURCENAME: KEGG GENOME
 | KEGGSOURCEURL: ftp://ftp.genome.jp/pub/kegg/genomes
 | KEGGSOURCEDATE: 2011-Mar15
 | GOSOURCENAME: Gene Ontology
 | GOSOURCEURL: http://current.geneontology.org/ontology/go-basic.obo
 | GOSOURCEDATE: 2020-09-10
 | EGSOURCEDATE: 2020-Sep23
 | EGSOURCENAME: Entrez Gene
 | EGSOURCEURL: ftp://ftp.ncbi.nlm.nih.gov/gene/DATA
 | ENSOURCEDATE: 2020-Aug18
 | ENSOURCENAME: Ensembl
 | ENSOURCEURL: ftp://ftp.ensembl.org/pub/current_fasta
 | UPSOURCENAME: Uniprot
 | UPSOURCEURL: http://www.UniProt.org/
 | UPSOURCEDATE: Mon Oct 5 00:25:28 2020
```

A TxDb package connects a set of genomic coordinates to various transcript oriented features.

In other words, TxDb packages provide gene annotation models

```
> TxDb.Scerevisiae.UCSC.sacCer3.sgdGene::TxDb.Scerevisiae.UCSC.sacCer3.sgdGene
TxDb object:
# Db type: TxDb
# Supporting package: GenomicFeatures
# Data source: UCSC
# Genome: sacCer3
# Organism: Saccharomyces cerevisiae
# Taxonomy ID: 4932
# UCSC Table: sgdGene
# Resource URL: http://genome.ucsc.edu/
# Type of Gene ID: Name of canonical transcript in cluster
# Full dataset: yes
# miRBase build ID: NA
# transcript_nrow: 6692
# exon_nrow: 7034
# cds_nrow: 7034
# Db created by: GenomicFeatures package from Bioconductor
# Creation time: 2015-10-07 18:20:42 +0000 (Wed, 07 Oct 2015)
# GenomicFeatures version at creation time: 1.21.30
# RSQLite version at creation time: 1.0.0
# DBSCHEMAVERSION: 1.1
```

TxDb databases can be explored with AnnotationDbi functions

```
> AnnotationDbi::keys(TxDb.Scerevisiae.UCSC.sacCer3.sgdGene, keytype = "GENEID") %>% glimpse
chr [1:6534] "Q0010" "Q0032" "Q0055" "Q0075" "Q0080" "Q0085" "Q0092" "Q0120" "Q0130" "Q0140" "Q0142" "Q0143"
> AnnotationDbi::keys(TxDb.Scerevisiae.UCSC.sacCer3.sgdGene, keytype = "TXNAME") %>% glimpse
chr [1:6692] "YAL069W" "YAL068W-A" "YAL067W-A" "YAL066W" "YAL064W-B" "YAL064W" "YAL062W" "YAL061W" "YAL060W"
```

Data packages that contain the full genome sequences of a given organism

```
> genome <- BSgenome.Scerevisiae.UCSC.sacCer3::BSgenome.Scerevisiae.UCSC.sacCer3
> genome
Yeast genome:
# organism: Saccharomyces cerevisiae (Yeast)
# genome: sacCer3
# provider: UCSC
# release date: April 2011
# 17 sequences:
# chrI chrII chrIII chrIV chrV chrVI chrVII chrVIII chrIX chrX chrXI chrXII chrXIII chrXIV chrXV chrXVI chrM
# (use 'seqnames()' to see all the sequence names, use the '$' or '[' operator to access a given sequence)
```

Biostrings package is used to interact with BSgenome databases

```
> genome <- BSgenome.Scerevisiae.UCSC.sacCer3::BSgenome.Scerevisiae.UCSC.sacCer3
> genome
Yeast genome:
# organism: Saccharomyces cerevisiae (Yeast)
# genome: sacCer3
# provider: UCSC
# release date: April 2011
# 17 sequences:
# chrI chrII chrIII chrIV chrV chrVI chrVII chrVIII chrIX chrX chrXI chrXII chrXIII chrXIV chrXV chrXVI chrM
# (use 'seqnames()' to see all the sequence names, use the '$' or '[' operator to access a given sequence)
> Biostrings::getSeq(genome)
DNAStringSet object of length 17:
      width seq
[1] 230218 CCACACCACACCCACACCCACACACCACACCACACACCACACCCACACCCCA...GTGTGGGTGTGGTGTGGTGTGGGTGTGGTGTGGGTGTGGTGTGTGGG chrI
[2] 813184 AAATAGCCCTCATGTACGTCTCCTCCAAGCCCTGTTGTCTCTTACCCGGATG...AGGGTGTGTGGGTGTGGGTGTGGTGTGTGTGGGTGTGGTGTGTGGGTGTGT chrII
[3] 316620 CCCACACACCACACCCACACCCACACCCACACACCACACACACCACACCCACA...GGTGTGGTGTGTGGGTGTGGGTGTGTGGGTGTGGTGGGTGTGGTGTGTGTG chrIII
[4] 1531933 ACACCACACCCACACCACACCCACACACACCACACCCACACACCACACCCAC...AGACAATCTATAAAAAGTAAACATAAAATAAAGGTAGTAAGTAGCTTTTGG chrIV
[5] 576874 CGTCTCCTCCAAGCCCTGTTGTCTCTTACCCGGATGTTCAACCAAAAAGCTAC...TAGAAGTGCTGTAGGGCTAAAGAACAGGGTTTCATTTTCATTTTTTTTTTT chrV
...
[13] 924431 CCACACACACACCACACCCACACCCACACCCACACCCACACCCACACCCACAC...TGGGTGTGGTGTGGTGTGTGGGTGTGGTGTGGGTGTGGTGTGTGTGTGGGG chrXIII
[14] 784333 CCGGCTTTCTGACCGAAATTAATAAAAAAAAAAATGAAATGAAACCCTGTTCT...GGGTGTGGTGTGTGGGTGTGGTGTGTGGGTGTGTGTGGGTGTGGTGTGGGT chrXIV
[15] 1091291 ACACCACACCCACACCACACCCACACCCACACACCACACCCACACACACACA...ATGACAGCGAGAGTAGAGGTAGATGTGAGAGAGTGTGTGGGTGTGGTGTGT chrXV
[16] 948066 AAATAGCCCTCATGTACGTCTCCTCCAAGCCCTGTTGTCTCTTACCCGGATG...CTAAAGAACAGGGTTTCATTTTCATTTTTTTTTTTTAAATTCGGTCAGAAA chrXVI
[17] 85779 TTCATAATTAATTTTTTATATATATATTATATTATAATATTAATTTATATTA...ATAGTTATATTATTATACAGAAATATGCTTAATTATAATATAATATCCATA chrM
```

AnnotationHub: retrieving release-specific files

The AnnotationHub package provides a client interface to resources stored at the AnnotationHub web service

It is different from AnnotationDbi-supported packages (e.g. orgDb or TxDb packages), since it allows access to files on top of databases

AnnotationHub: retrieving release-specific files

The AnnotationHub package provides a client interface to resources stored at the AnnotationHub web service.

```
> ah <- AnnotationHub::AnnotationHub()
snapshotDate(): 2020-10-27
> ah
AnnotationHub with 54989 records
# snapshotDate(): 2020-10-27
# $dataprovder: Ensembl, BroadInstitute, UCSC, ftp://ftp.ncbi.nlm.nih.gov/gene/DATA/, Haemcode, FungiDB, Inparanoid8, TriTrypDB, Plasmo...
# $species: Homo sapiens, Mus musculus, Drosophila melanogaster, Bos taurus, Pan troglodytes, Rattus norvegicus, Danio rerio, Gallus gal...
# $rdataclass: GRanges, TwoBitFile, BigWigFile, EnsDb, Rle, OrgDb, ChainFile, TxDb, Inparanoid8Db, data.frame
# additional mcols(): taxonomyid, genome, description, coordinate_1_based, maintainer, rdatadateadded, preparerclass, tags,
#   rdatapath, sourceurl, sourcetype
# retrieve records with, e.g., 'object[["AH5012"]]'

      title
AH5012 | Chromosome Band
AH5013 | STS Markers
AH5014 | FISH Clones
AH5015 | Recomb Rate
AH5016 | ENCODE Pilot
...
AH89321 | Ensembl 102 EnsDb for Xiphophorus couchianus
AH89322 | Ensembl 102 EnsDb for Xiphophorus maculatus
AH89323 | Ensembl 102 EnsDb for Xenopus tropicalis
AH89324 | Ensembl 102 EnsDb for Zonotrichia albicollis
AH89325 | Ensembl 102 EnsDb for Zalophus californianus
```

- Queries are done
using query(ah,
“keyword”)

AnnotationHub: retrieving release-specific files

Queries are done using `query(ah, c("sacCer3", "TwoBitFile"))`

```
> query(ah, c('sacCer3', 'TwoBitFile'))
AnnotationHub with 1 record
# snapshotDate(): 2020-10-27
# names(): AH14104
# $dataprovder: UCSC
# $species: Saccharomyces cerevisiae
# $rdataclass: TwoBitFile
# $rdatadateadded: 2014-12-15
# $title: sacCer3.2bit
# $description: UCSC 2 bit file for sacCer3
# $taxonomyid: 4932
# $genome: sacCer3
# $sourcetype: TwoBit
# $sourceurl: http://hgdownload.cse.ucsc.edu/goldenpath/sacCer3/bigZips/sacCer3.2bit
# $sourcesize: NA
# $tags: c("2bit", "UCSC", "genome")
# retrieve record with 'object[["AH14104"]]'
```

AnnotationHub: retrieving release-specific files

Queries are done using `query(ah,`

Objects are retrieved using `ah["`.

```
> query(ah, c('sacCer3', 'TwoBitFile'))
AnnotationHub with 1 record
# snapshotDate(): 2020-10-27
# names(): AH14104
# $dataprovder: UCSC
# $species: Saccharomyces cerevisiae
# $rdataclass: TwoBitFile
# $rdatadateadded: 2014-12-15
# $title: sacCer3.2bit
# $description: UCSC 2 bit file for sacCer3
# $taxonomyid: 4932
# $genome: sacCer3
# $sourcetype: TwoBit
# $sourceurl: http://hgdownload.cse.ucsc.edu/goldenpath/sacCer3/bigZips/sacCer3.2bit
# $sourcesize: NA
# $tags: c("2bit", "UCSC", "genome")
# retrieve record with 'object[["AH14104"]]'
> twobit <- ah[['AH14104']]
```

loading from cache

```
> twobit
```

TwoBitFile object

```
resource: /Users/jacquesserizay/Library/Caches/AnnotationHub/a6c5475f3d0f_18199
```

```
> seqs <- getSeq(twobit)
```

```
> seqs
```

DNASet object of length 17:

	width	seq	names
[1]	230218	CCACACCACACCCACACCCACACACCACACCACACACCACACCACACCCA...GTGTGGTGTGGTGTGGTGTGGGTGTGGGTGTGGGTGTGGTGTGTGGG	chrI
[2]	813184	AAATAGCCCTCATGTACGCTCTCTCAAGCCCTGTTGTCTTTACCCGGATG...AGGGTGTGTGGGTGTGGGTGTGGTGTGTGGGTGTGGTGTGGGTGTGT	chrII
[3]	316620	CCCACACACCACACCCACACCACCCACACACCACACACCACACCCACAC...GGTGTGGTGTGTGGGTGTGGGTGTGTGGGTGTGGTGGGTGTGGTGTGTGTG	chrIII
[4]	1531933	ACACCACACCCACACCACACCCACACACCCACACCCACACACCACACCCAC...AGACAATCTATAAAAAAGTAAACATAAAATAAAGGTAGTAAGTAGCTTTTGG	chrIV
[5]	439888	CACACACACCACACCCACACCACACCACACCACACCCACACCCACACACACC...GGATGTGGTGTGGATGTGGTGTGGGTGTGGATGTGGGTGTGGTGTGTGTGT	chrIX
...	...	...	
[13]	924431	CCACACACACACCACACCCACACCACACCCACACCCACACCCACACCCACAC...TGGGTGTGGTGTGGTGTGTGGGTGTGGTGTGGGTGTGGTGTGTGTGTGGGG	chrXIII
[14]	784333	CCGGCTTTCTGACCGAAATTAATAAAAAAAAAAATGAAATGAAACCTGTCT...GGGTGTGGTGTGTGGGTGTGGTGTGTGGGTGTGTGTGGGTGTGGTGTGGGT	chrXIV
[15]	1091291	ACACCACACCCACACCACACCCACACCCACACACCACACCCACACACACAC...ATGACAGCGAGAGTAGAGGTAGATGTGAGAGAGTGTGTGGGTGTGGTGTGT	chrXV
[16]	948066	AAATAGCCCTCATGTACGCTCTCTCAAGCCCTGTTGTCTTTACCCGGATG...CTAAAGAACAGGGTTTCATTTTCATTTTTTTTTTTTAAATTTGGTCAGAAA	chrXVI
[17]	85779	TTCAATAATTAATTTTTTATATATATATTATATTATAATTAATTTATATTA...ATAGTTATATTATTATACAGAAATATGCTTAATTATAATATAATATCCATA	chrM

AnnotationHub: retrieving release-specific files

Many **many** resources available on AnnotationHub

AnnotationHub: retrieving release-specific files

```
> query(ah, 'VcfFile')
AnnotationHub with 8 records
# snapshotDate(): 2020-10-27
# $dataprovder: dbSNP
# $species: Homo sapiens
# $rdataclass: VcfFile
# additional mcols(): taxonomyid, genome, description, coordinate_1_based, maintainer, rdatadateadded, preparerclass, tags,
#   rdatapath, sourceurl, sourcetype
# retrieve records with, e.g., 'object[["AH57956"]]'
```

	title
AH57956	clinvar_20160203.vcf.gz
AH57957	clinvar_20160203_papu.vcf.gz
AH57958	common_and_clinical_20160203.vcf.gz
AH57959	common_no_known_medical_impact_20160203.vcf.gz
AH57960	clinvar_20160203.vcf.gz
AH57961	clinvar_20160203_papu.vcf.gz
AH57962	common_and_clinical_20160203.vcf.gz
AH57963	common_no_known_medical_impact_20160203.vcf.gz

AnnotationHub: retrieving release-specific files

```
> query(ah, c('bigwig', 'UCSC') )
AnnotationHub with 2198 records
# snapshotDate(): 2020-10-27
# $dataprovder: UCSC
# $species: Homo sapiens, Drosophila melanogaster, Mus musculus
# $rdataclass: Rle, GRanges
# additional mcols(): taxonomyid, genome, description, coordinate_1_based, maintainer, rdatadateadded, preparerclass, tags,
#   rdatapath, sourceurl, sourcetype
# retrieve records with, e.g., 'object[["AH23256"]]'

      title
AH23256 | wgEncodeBroadHistoneGm12878H3k4me3StdPk.broadPeak.gz
AH23257 | wgEncodeBroadHistoneGm12878H3k9acStdPk.broadPeak.gz
AH23262 | wgEncodeBroadHistoneGm12878H3k36me3StdPk.broadPeak.gz
AH23367 | wgEncodeBroadHistoneHuvecH3k27me3StdPk.broadPeak.gz
AH24345 | wgEncodeCshlLongRnaSeqNhemfm2CellTotalGeneGencV10.gtf.gz
...
AH78698 | phastCons30way.UCSC.hg38.chrX.rds
AH78699 | phastCons30way.UCSC.hg38.chrX_KI270880v1_alt.rds
AH78700 | phastCons30way.UCSC.hg38.chrX_KI270881v1_alt.rds
AH78701 | phastCons30way.UCSC.hg38.chrX_KI270913v1_alt.rds
AH78702 | phastCons30way.UCSC.hg38.chrY.rds
```

AnnotationHub: retrieving release-specific files

```
> query(ah, c('TxDb', 'GENCODE'))
AnnotationHub with 20 records
# snapshotDate(): 2020-10-27
# $dataprovder: GENCODE
# $species: Homo sapiens
# $rdataclass: TxDb
# additional mcols(): taxonomyid, genome, description, coordinate_1_based, maintainer, rdatadateadded, preparerclass, tags,
#   rdatapath, sourceurl, sourcetype
# retrieve records with, e.g., 'object[["AH75134"]]'
```

	title
AH75134	TxDb for Gencode v23 on hg19 coordinates
AH75137	TxDb for Gencode v23 on hg38 coordinates
AH75140	TxDb for Gencode v24 on hg19 coordinates
AH75143	TxDb for Gencode v24 on hg38 coordinates
AH75146	TxDb for Gencode v25 on hg19 coordinates
...	...
AH75179	TxDb for Gencode v30 on hg38 coordinates
AH75182	TxDb for Gencode v31 on hg19 coordinates
AH75185	TxDb for Gencode v31 on hg38 coordinates
AH75188	TxDb for Gencode v32 on hg19 coordinates
AH75191	TxDb for Gencode v32 on hg38 coordinates